

Prompt 4: Initial concept-describing prompt used in construction of Bongard-RWR.

Your goal is to translate a comparison concept from the geometric domain to the
 ↪ real-world domain. Your translations should be expressible as images.

Example:

Geometric domain: triangles vs squares

```
{
  "left": {
    "concept": "pyramids"
  },
  "right": {
    "concept": "rectangular buildings"
  }
}
```

Give <number> unique translations for the following concept as a raw JSON array of
 ↪ objects (same as in the example above).
 <concept>

Prompt 5: Prompt used for the selection of proper images for a translated concept.

You translated a concept comparison from geometric domain to the real-world domain as
 ↪ follows:

Geometric domain: <left_geometric_concept> vs <right_geometric_concept>

Real world domain:

```
{
  "left": {
    "concept": <left_concept>,
    "prompt": <left_concept_description>
  },
  "right": {
    "concept": <right_concept>,
    "prompt": <right_concept_description>
  }
}
```

Now, you need to check if the queried image matches your translation and provides enough
 ↪ information to distinguish it from the other concept. Don't focus too much on the
 ↪ prompt. It's just a hint for you to understand the concept better.
 Provided image represents <side_concept>

Give your answer in the following format:

EVALUATION: OK

EXPLANATION: <here you can provide additional information>

or

EVALUATION: REJECTED

EXPLANATION: <here you can provide additional information>

Prompt 6: Prompt explaining Bongard problems to an MLLM.

A Bongard Problem is composed of left and right sides separated by a line. Each side
 ↪ contains six images. All images belonging to one side present a common concept,
 ↪ which is lacking in all images from the other side. The goal is to describe the rule
 ↪ that fits all images on the left side, but none on the right, and, conversely, the
 ↪ rule that fits all images on the right side, but none on the left. The description
 ↪ of the rule should be simple and concise.

Example 1: All shapes on left are small. All shapes on right are big.

Example 2: The left side contains circles. The right side contains triangles.

Prompt 7: Prompt used for images to side classification (see Fig. 3c). Two examples were provided to not bias the results of the model.

You are a vision understanding module designed to provide short, clear and accurate
 ↳ answers. Your goal is to classify two test images to the corresponding side of the
 ↳ Bongard Problem, LEFT or RIGHT. Each image belongs to exactly one class. The test
 ↳ images belong to different classes.

The images are always provided correctly. Respond only to the specific request. Respond
 ↳ in json using the following format.

FIRST EXAMPLE:

Left images: <small shapes>

Right images: <big shapes>

First test image: <small shape>

Second test image: <big shape>

Response:

```
{
  "first": {
    "explanation": "The test image shows a small shape, similarly as all images on
  ↳ the left side. Conversely, the images on the right side feature big
  ↳ shapes.",
    "concept": "small vs big",
    "answer": "LEFT"
  },
  "second": {
    "explanation": "The test image shows a big shape, similarly as all images on
  ↳ right. The images on left, on the other hand, feature small shapes.",
    "concept": "small vs big",
    "answer": "RIGHT"
  }
}
```

END OF FIRST EXAMPLE

SECOND EXAMPLE:

Left images: <circles>

Right images: <triangles>

First test image: <triangle>

Second test image: <circle>

Response:

```
{
  "first": {
    "explanation": "The test image shows a triangle, which matches all images on
  ↳ right. In contrast, the left side images feature circles.",
    "concept": "circles vs triangles",
    "answer": "RIGHT"
  },
  "second": {
    "explanation": "The test image shows a circle, which matches all images on left.
  ↳ Conversely, the right side images feature triangles.",
    "concept": "circles vs triangles",
    "answer": "LEFT"
  }
}
```

END OF SECOND EXAMPLE

Prompt 8: Prompt used for the solution correctness assessment (see Fig. 3b).

You are a vision understanding module designed to provide short, clear and accurate
 ↳ answers. Your goal is to evaluate the correctness of the provided answer to the
 ↳ given Bongard Problem. All images are provided correctly. Do not explain the answer,
 ↳ just evaluate it. Respond 'OK' if the answer is correct, otherwise respond 'WRONG'.

User answer: <user_answer>

Prompt 9: Prompt used for the *Direct* strategy. (see Fig. 2a).

You are a vision understanding module designed to provide short, clear and accurate
→ answers. Your goal is to solve the provided Bongard Problem. What is the difference
→ between the two sides of the problem?

Prompt 10: Prompt used to obtain the image descriptions in the *Descriptive* strategy (see Fig. 2b).

The provided image is a part of an abstract visual reasoning problem. Describe all
→ crucial properties of the image. Your description should be as concise as possible.
→ Focus on the most important details. The image is provided correctly. Respond only
→ with descriptions.

I.4. Prompts for Natural Language Answer Generation Strategies

Prompts 9–16 were used for natural language answer generation (see Section 3.1).

Prompt 11: Prompt used for the *Descriptive* and *Descriptive-direct* strategies (see Fig. 2b).

You are a vision understanding module designed to provide short, clear and accurate
→ answers. Your goal is to solve the provided Bongard Problem using descriptions of
→ its images.

LEFT IMAGES:
<left_descriptions>

RIGHT IMAGES:
<right_descriptions>

What is the difference between the two sides of the problem?